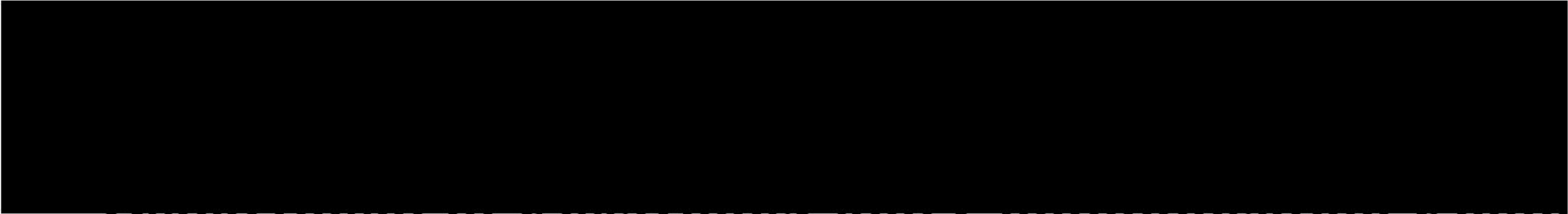
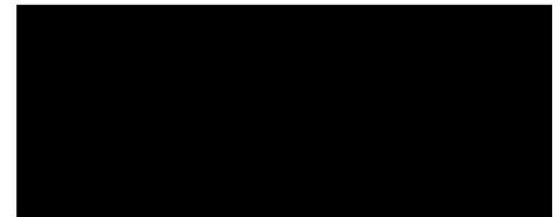




Unit-I



**Department of Electronics and Communication Engineering
Amrita School of Engineering, Chennai campus**



Unit I :

Introduction: major blocks and functions; Analog to Digital Conversion; Waveform Coding –PCM – DPCM –DM; Time Division Multiplexing; Geometric representation of signal waveforms- Binary pulse modulation –Optimum receiver for binary modulated signals in additive white Gaussian noise: M-ary binary and orthogonal pulse modulation –Probability of error for binary and M-ary pulse modulation.



Basic Block Diagram of Digital Communication System

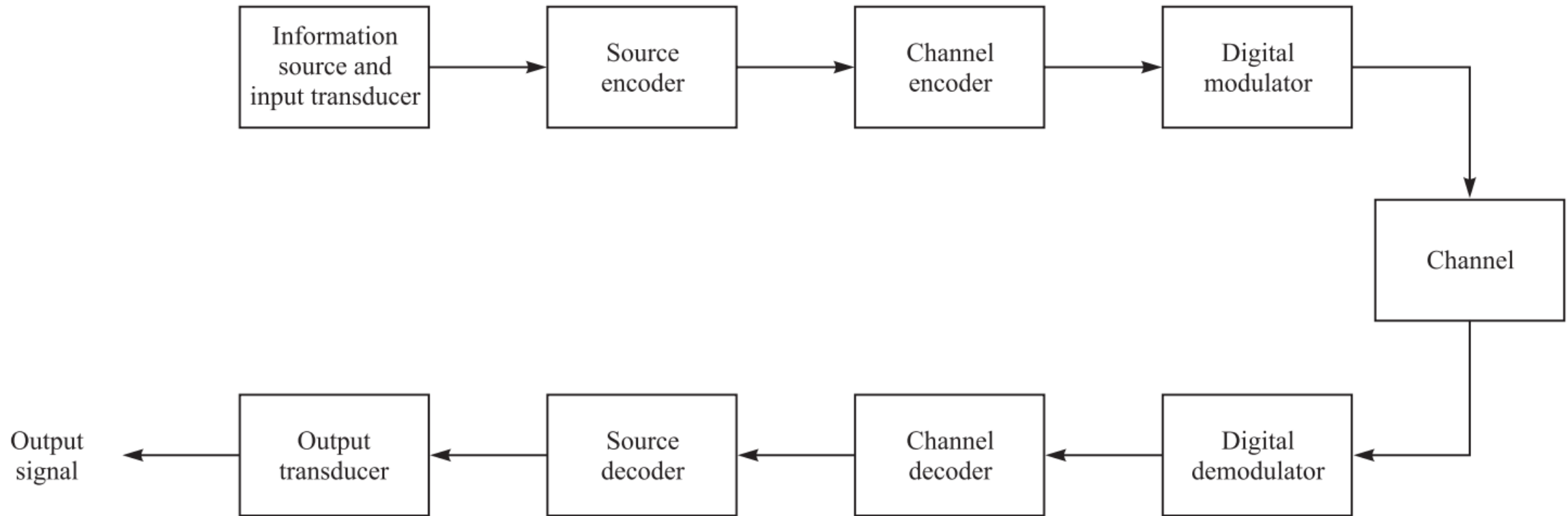


FIGURE 1.1-1

Basic elements of a digital communication system.



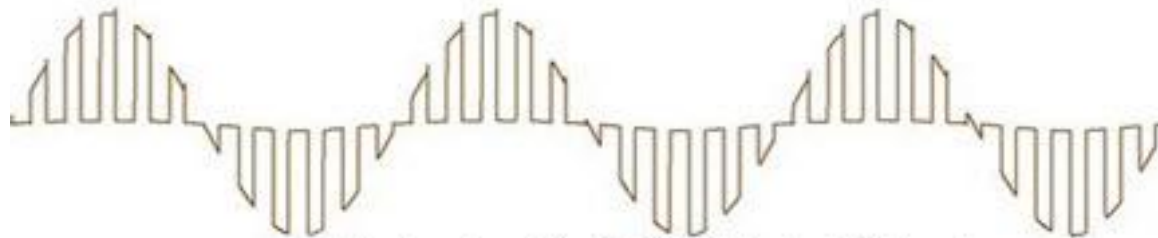
Pulse Analog Modulation Techniques



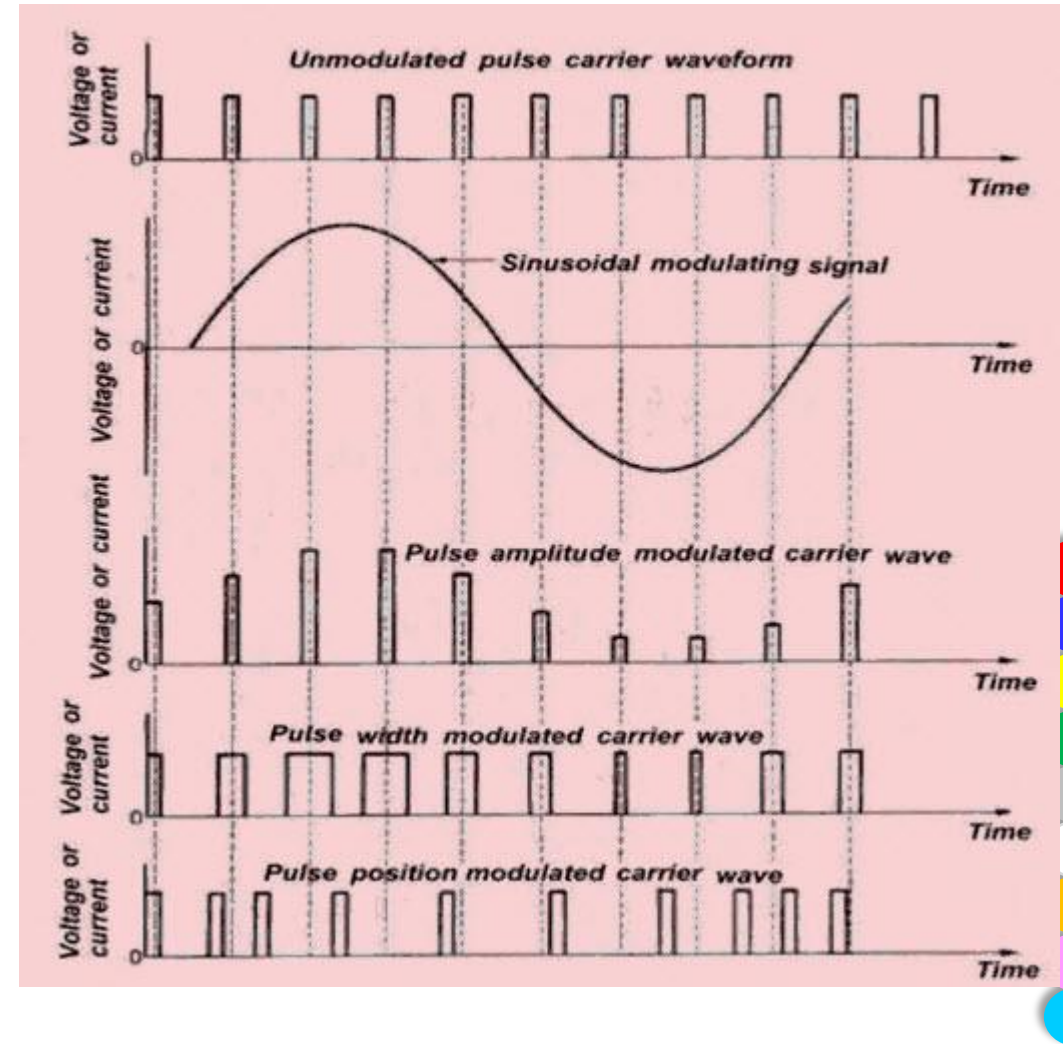
Carrier Pulses Train Used for Modulation



Analog Signal for Modulating



Pulse Amplitude Modulated Output



	PAM	PWM/PDM	PPM
1)	Waveform	Waveform	Waveform
2)	Amplitude of the pulse is proportional to amplitude of modulating signal	Width of the pulse is proportional to amplitude of modulating signal	The relative position of the pulse is proportional to amplitude of modulating signal
3)	Width constant position to amplitude of modulating signal	Amplitude constant position to amplitude of modulating signal	Amplitude constant width to amplitude of modulating signal
4)	The B.W of the transmission Channel depends on width of the pulse	B.W of transmission channel depends on rise time of the pulse	B.W of transmission channel depends on rising time of the pulse
5)	The instantaneous power of transmitter varies	The instantaneous power of the transmitter varies	The instantaneous power of the transmitter remains constant.
6)	Noise interference is high	Noise interference is minimum	Noise interference is minimum
7)	System is complex similar to Amplitude modulation	Simple to implement similar to frequency Modulation	Simple to implement similar to phase. modulation



Sampling Process

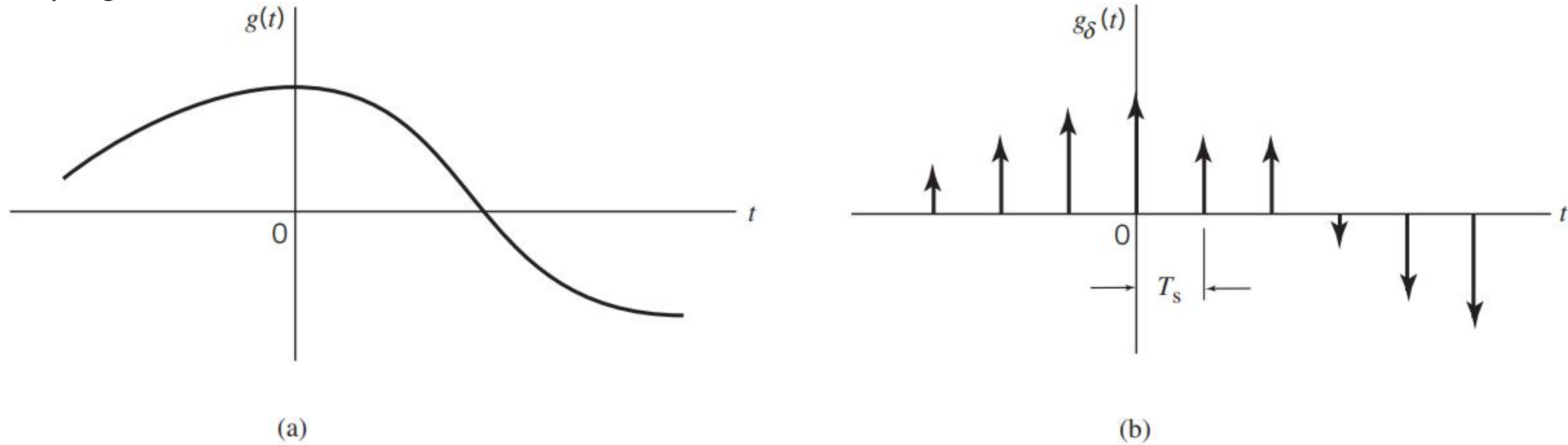


Figure 6.1 The sampling process. (a) Analog signal. (b) Instantaneously sampled version of the analog signal.

$$g_{\delta}(t) = \sum_{n=-\infty}^{\infty} g(nT_s) \delta(t - nT_s)$$



$$g_{\delta}(t) \rightleftharpoons_{f_s} \sum_{m=-\infty}^{\infty} G(f - mf_s) \quad (6.2)$$

where $G(f)$ is the Fourier transform of the original signal $g(t)$ and f_s is the sampling rate.

$$G_{\delta}(f) = \sum_{n=-\infty}^{\infty} g(nT_s) \exp(-j2\pi nfT_s)$$



Critical Sampling

1. $G(f) = 0$ for $|f| \geq W$.
2. $f_s = 2W$.

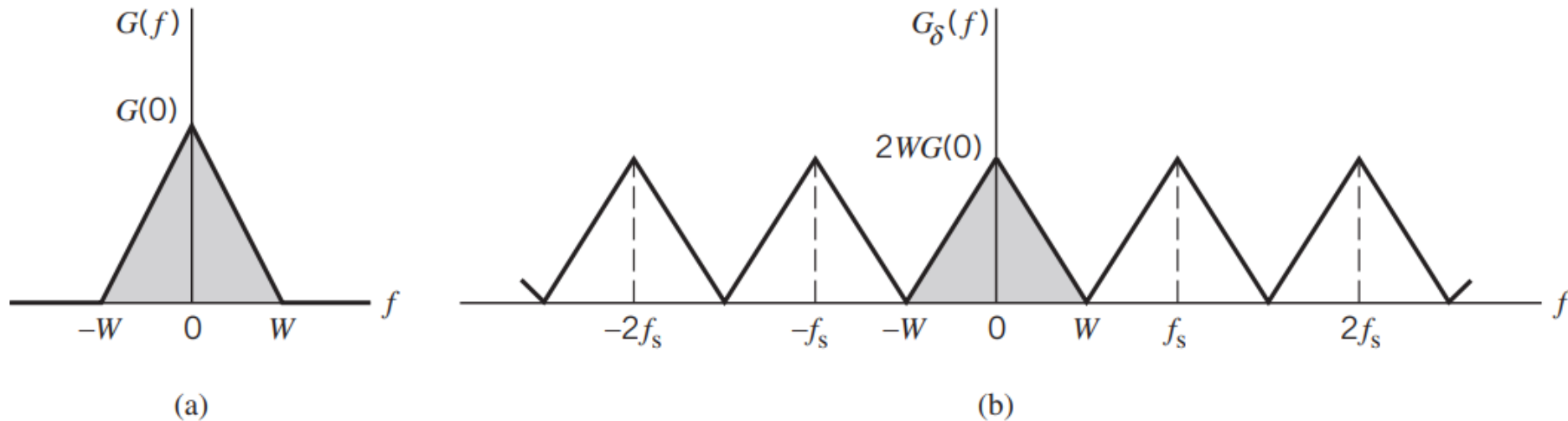


Figure 6.2 (a) Spectrum of a strictly band-limited signal $g(t)$. (b) Spectrum of the sampled version of $g(t)$ for a sampling period $T_s = 1/2W$.



Under Sampling

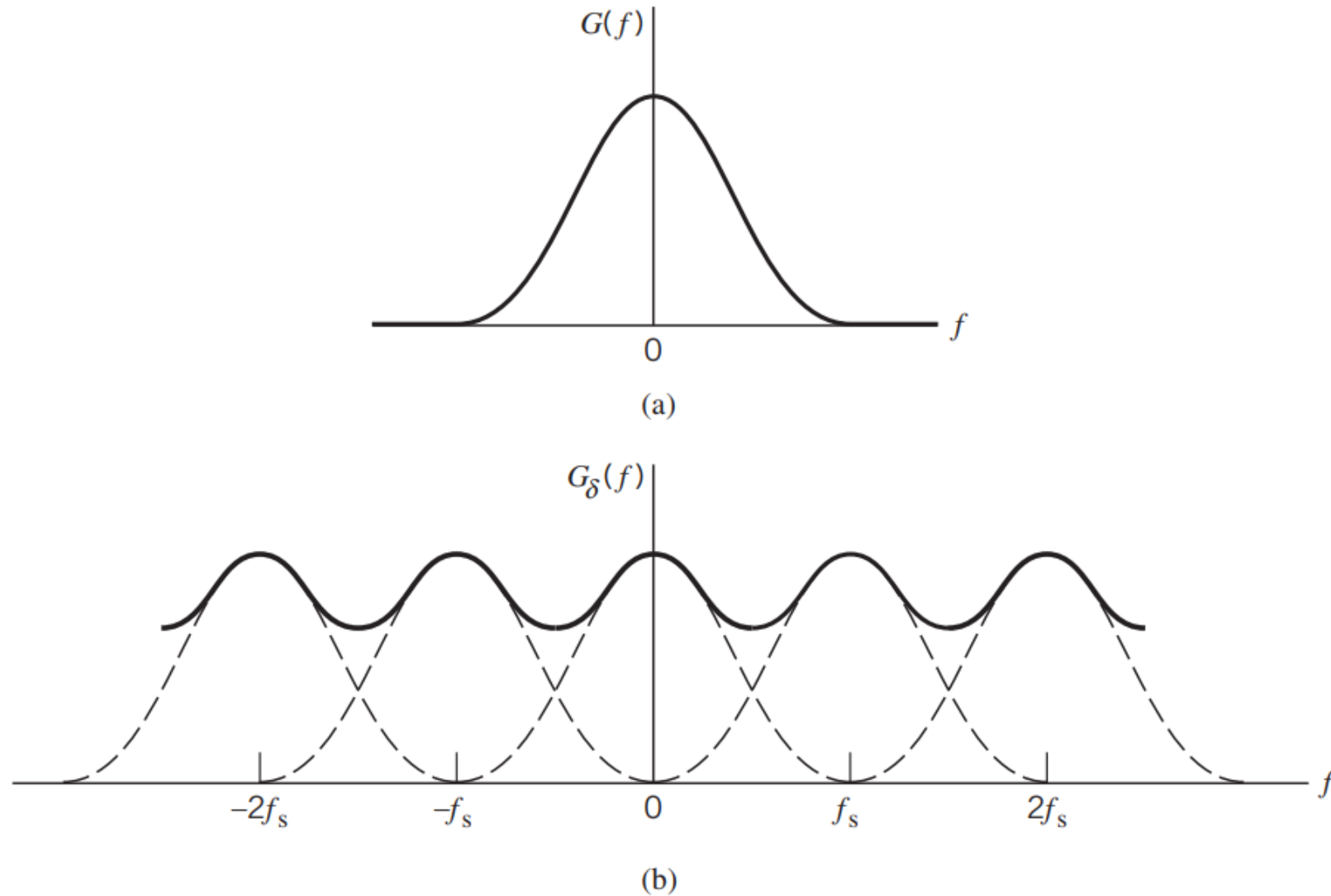


Figure 6.3 (a) Spectrum of a signal. (b) Spectrum of an under-sampled version of the signal exhibiting the aliasing phenomenon.



Over Sampling

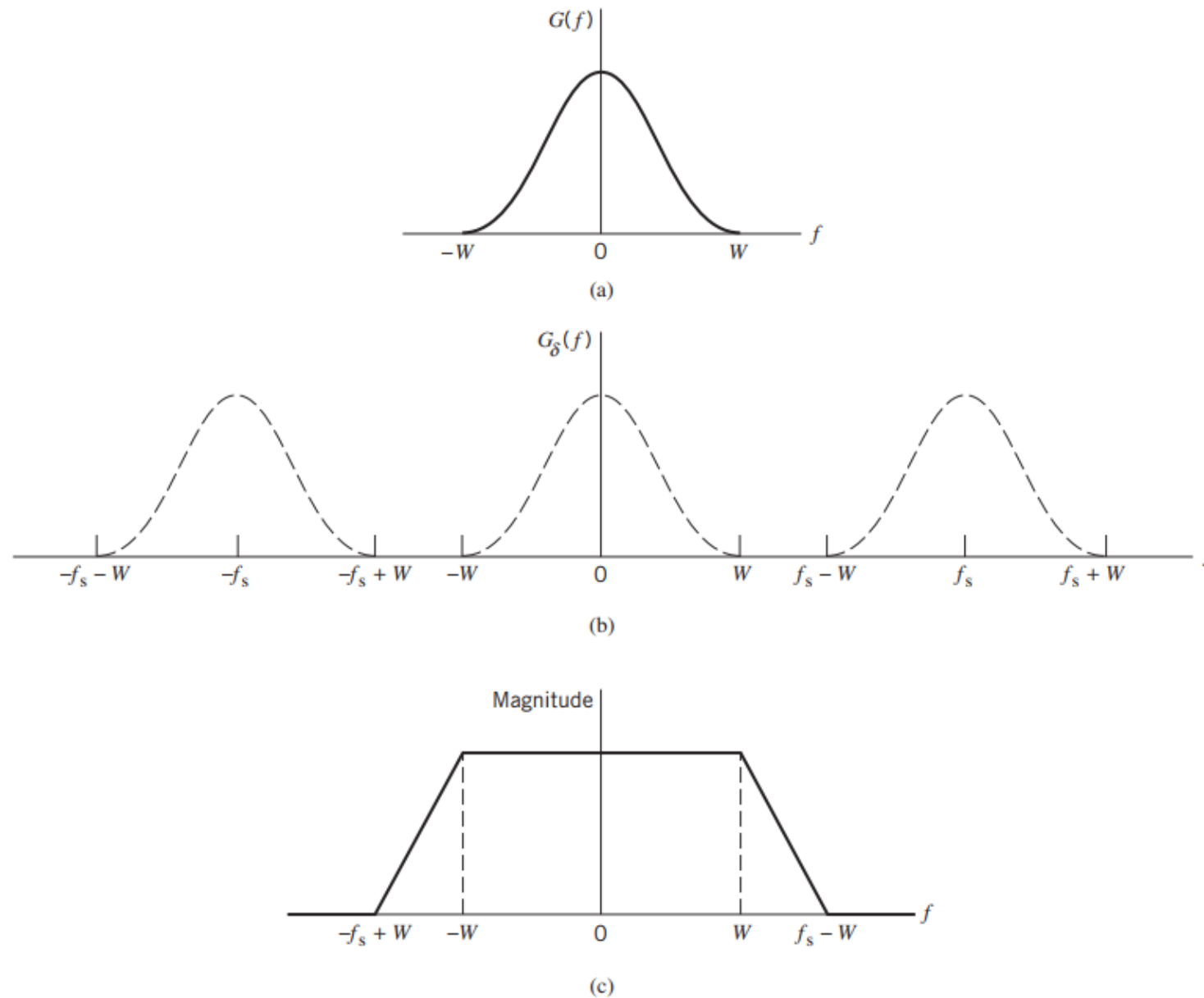


Figure 6.4 (a) Anti-alias filtered spectrum of an information-bearing signal. (b) Spectrum of instantaneously sampled version of the signal, assuming the use of a sampling rate greater than the Nyquist rate. (c) Magnitude response of reconstruction filter.



Find the Nyquist rate and the Nyquist interval for

(a) $m(t) = 5 \cos 1000\pi t \cos 4000\pi t$

(b) $m(t) = \frac{\sin 200\pi t}{\pi t}$

(c) $m(t) = \left(\frac{\sin 200\pi t}{\pi t} \right)^2$



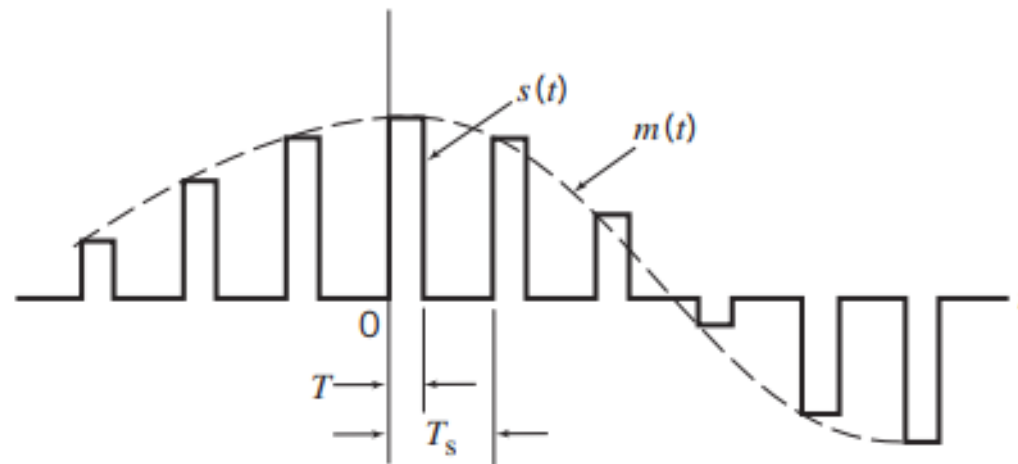
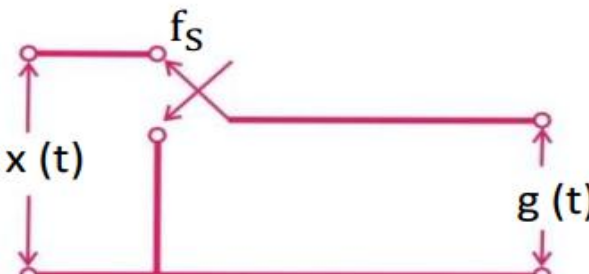
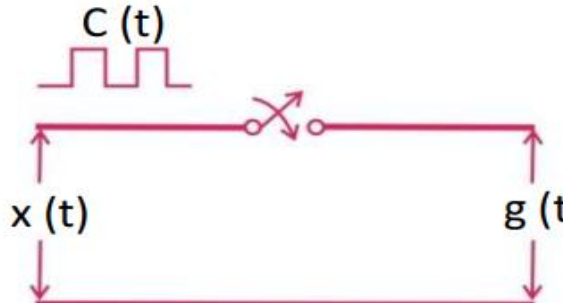
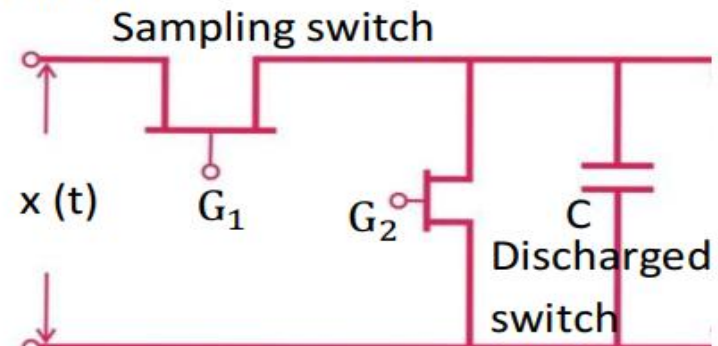
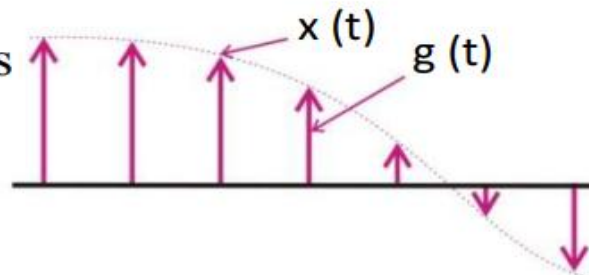
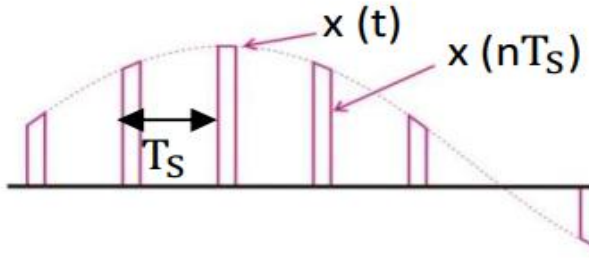
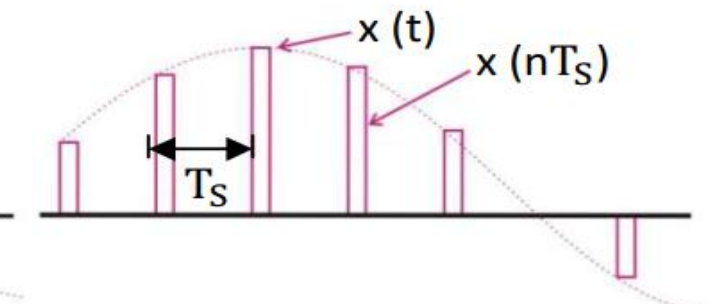


Figure 6.5 Flat-top samples, representing an analog signal.



	Parameter	Ideal/Instantaneous Sampling	Natural Sampling	Flat top Sampling
1)	Sampling Principle	It uses multiplication	It was chopping Principle	It uses sample and hold circuit
2)	Generation circuit			
3)	Wave forms			
4)	Feasibility	This is not a practically possible method	This method is used practically	This method is also used practically
5)	Sampling Rate criteria	It tends to infinity	Sampling rate satisfies Nyquist criteria	Sampling rate Satisfies Nyquist criteria

A signal $x_1(t)$ is band limited to 2KHz while $x_2(t)$ is band limited to 3KHz. Find the Nyquist sampling rate for

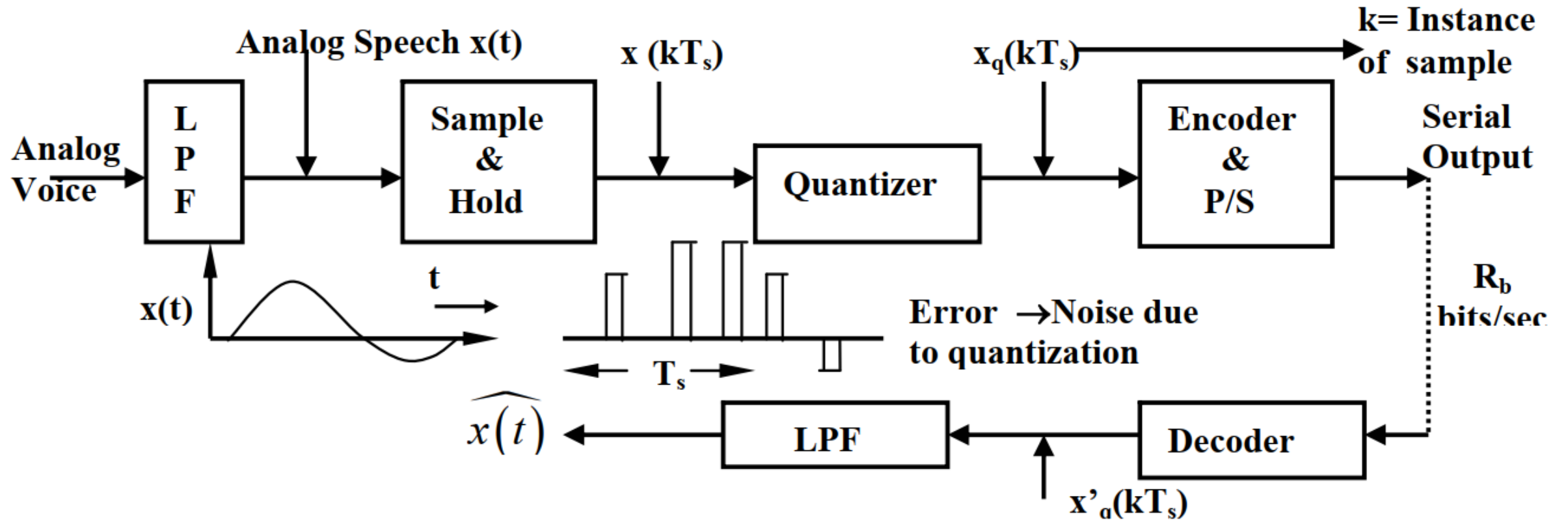
- a) $x_1(2t)$ b) $x_2(t-3)$ c) $x_1(t) + x_2(t)$ d) $x_1(t) \cdot x_2(t)$ e) $x_1(t) * x_2(t)$





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Schematic Diagram of Pulse Code Modulation



In a binary PCM system, the output signal-to-quantizing-noise ratio is to be held to a minimum of 40 dB. Determine the number of required levels, and find the corresponding output signal-to-quantizing-noise ratio.

The peak-to-peak input to an 8-bit PCM coder is 2 volts. The signal power-to-quantization noise power ratio (in dB) for an input of $0.5\cos(\omega_m t)$ is **(GATE-1999)**

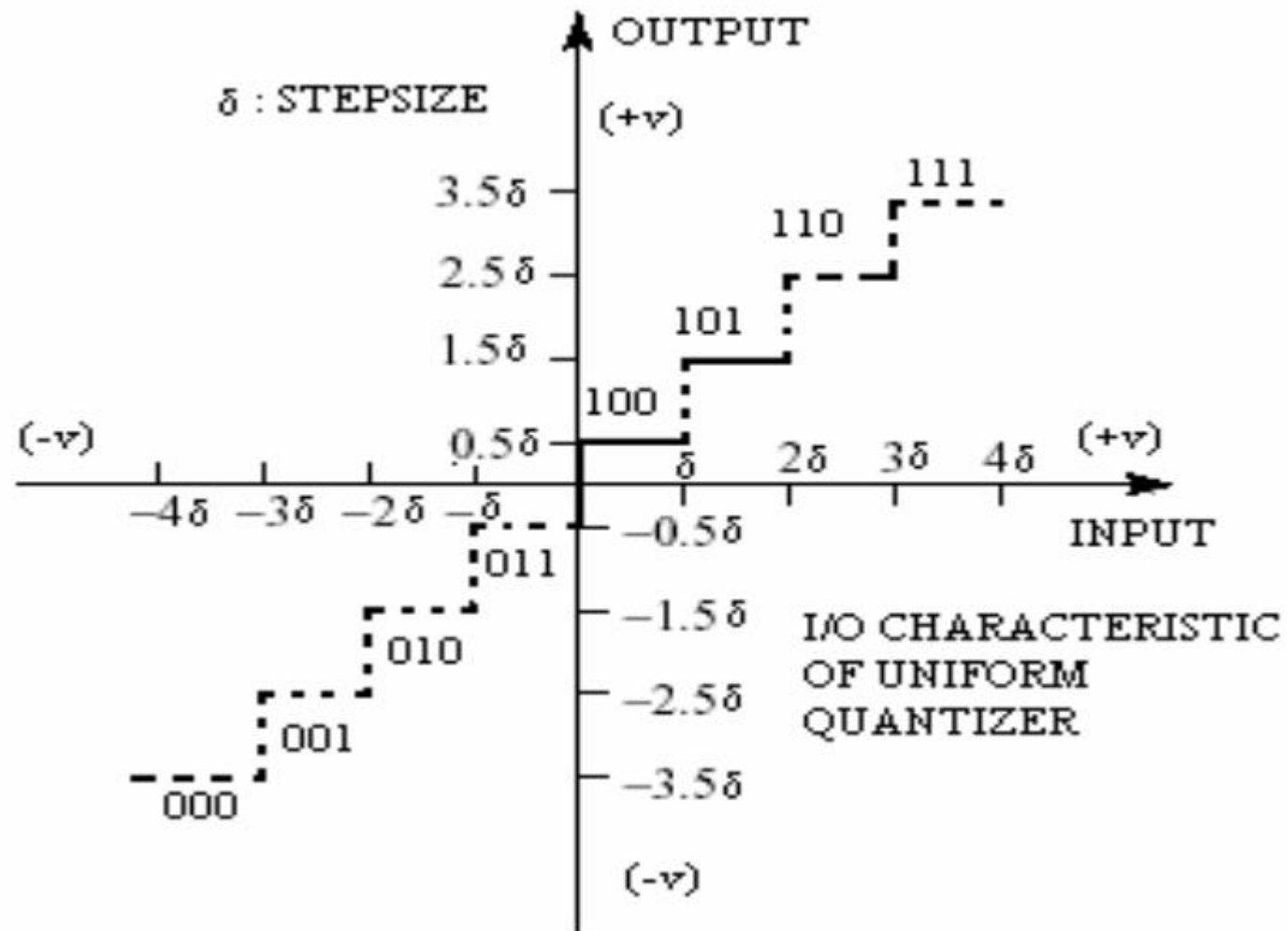
- a) 47.8 b) 43.8 c) 95.6 d) 99.6

A video transmission system transmits 625 picture frames per second. Each frame consists of a 400X400 pixel grid with 64 intensity levels per pixel. The data rate of the system is **(GATE-2001)**

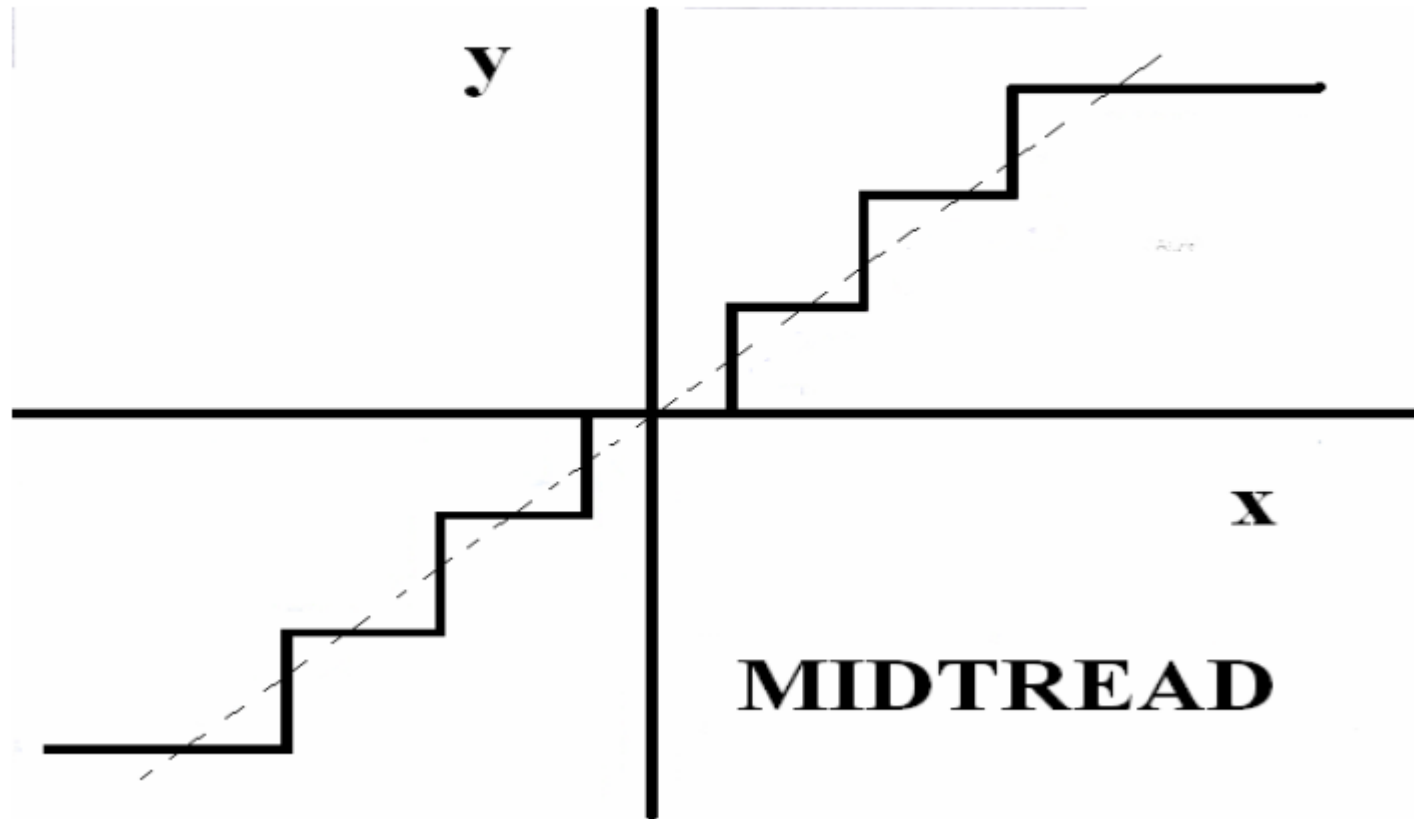
- a) 16 Mbps b) 100Mbps c) 600Mbps d) 6.4 Gbps



Uniform Quantizer: Mid-riser Quantization



Uniform Quantizer: Mid-tread Quantization



Non-Uniform Quantizer

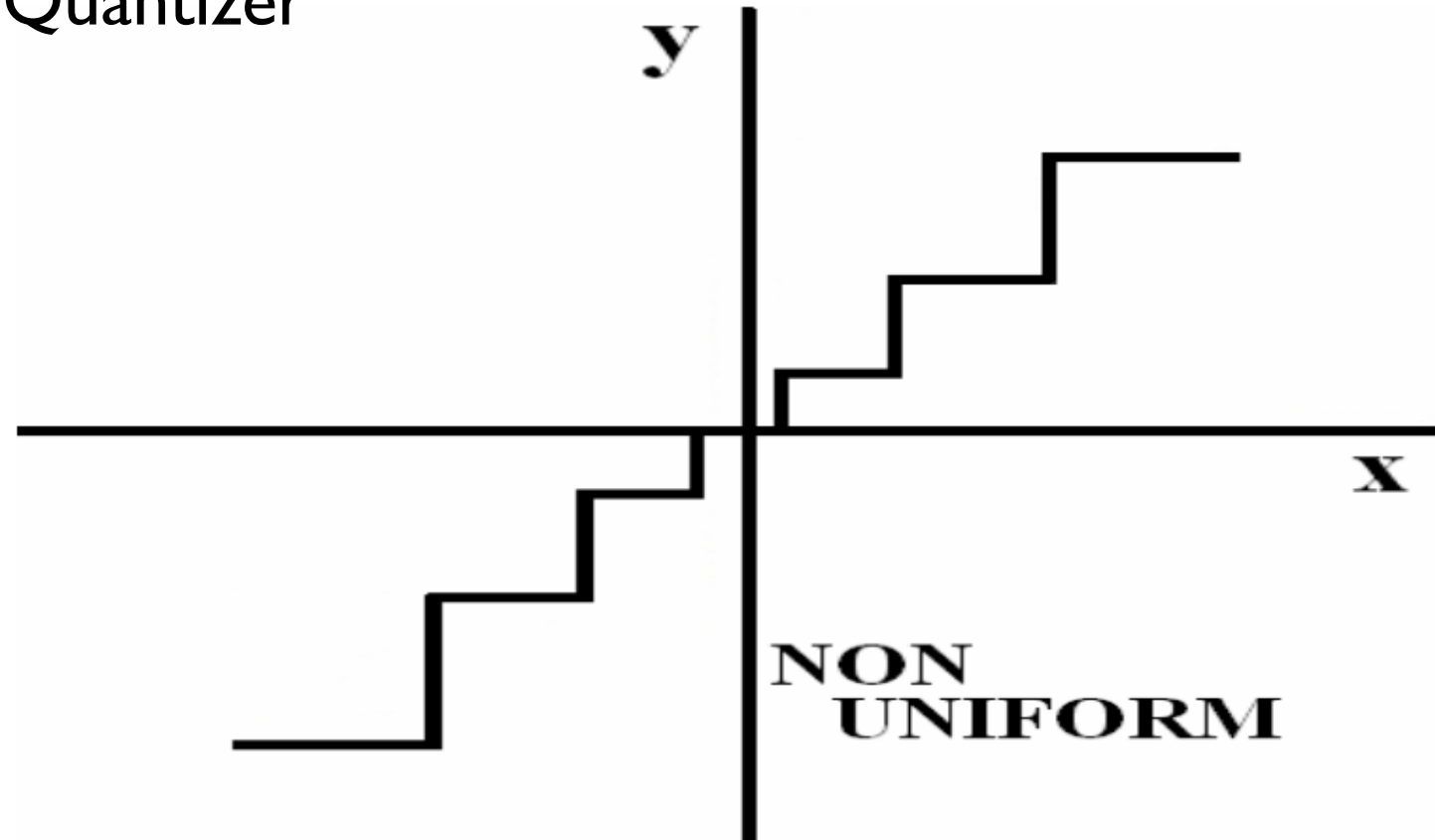


Fig 3.12.1 *Transfer characteristic of a non-uniform quantizer*



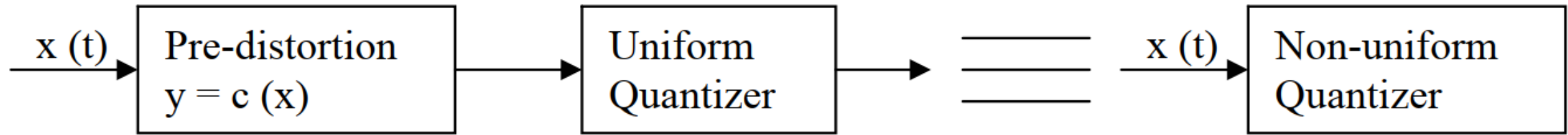


Fig. 3.12.2 *An equivalent form of a non-uniform quantizer*

There are two popular standards for non-linear quantization known as

- (a) The μ - law companding
- (b) The A – law companding.



μ - law companding

$$\frac{c(|x|)}{V} = \frac{\ln\left(1 + \frac{\mu|x|}{V}\right)}{\ln(1 + \mu)}, \quad 0 \leq \frac{|x|}{V} \leq 1.0$$

μ - a constant - 0 to 255
0 – linear quantization



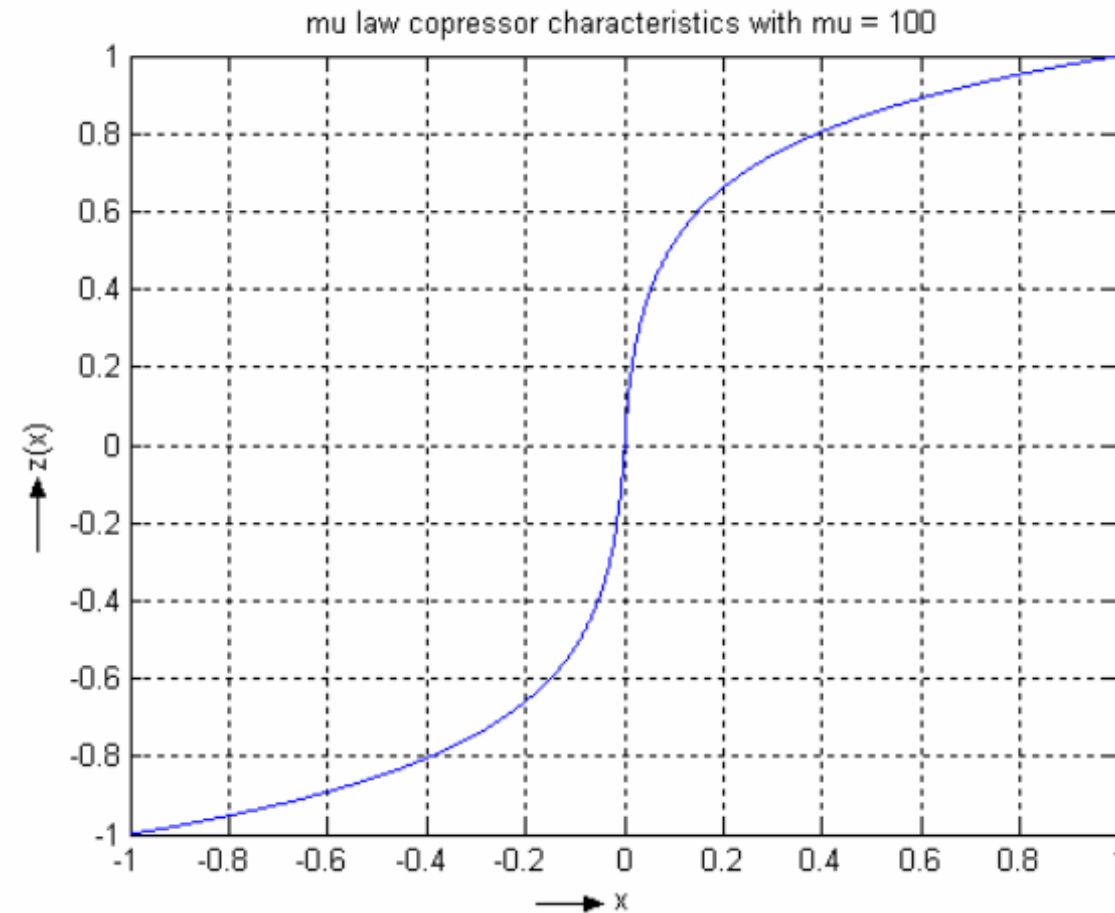
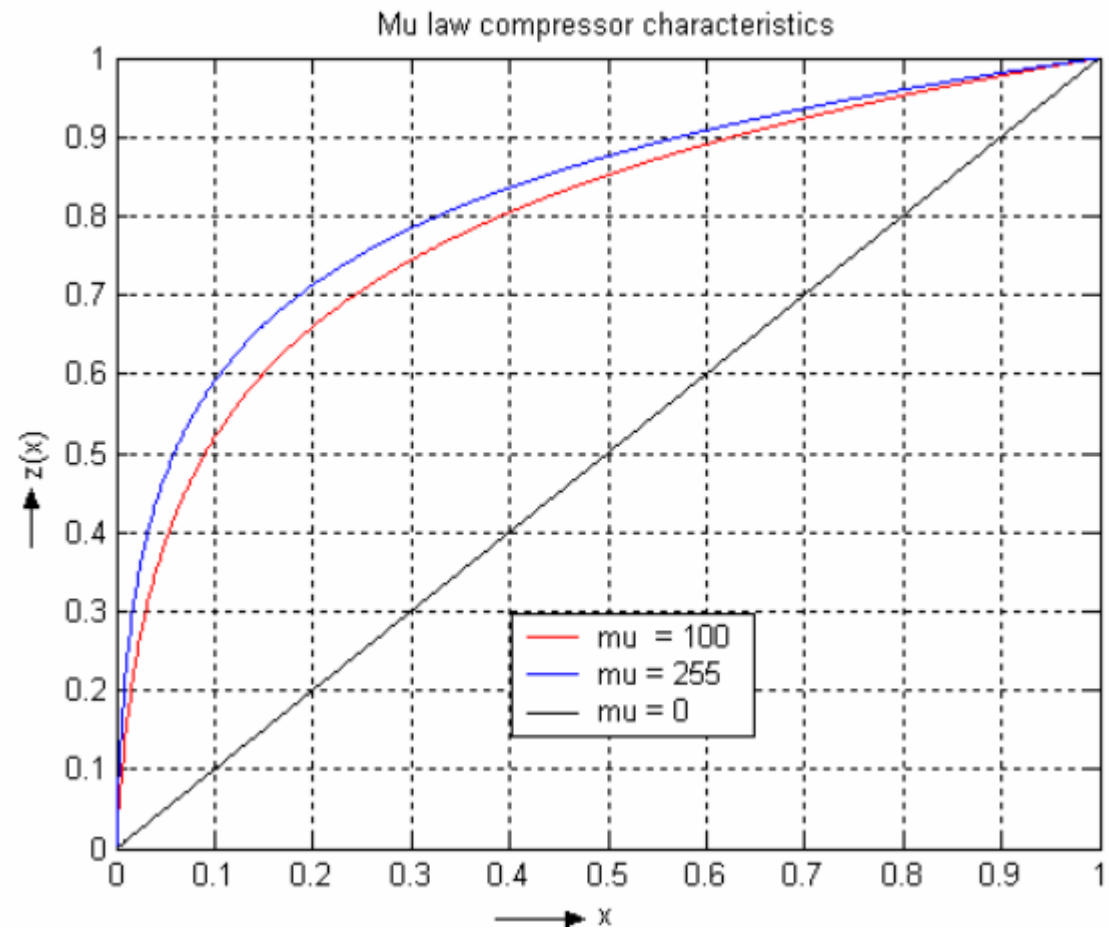


Fig. 3.12.4 μ -law companding characteristics($\mu = 100$)



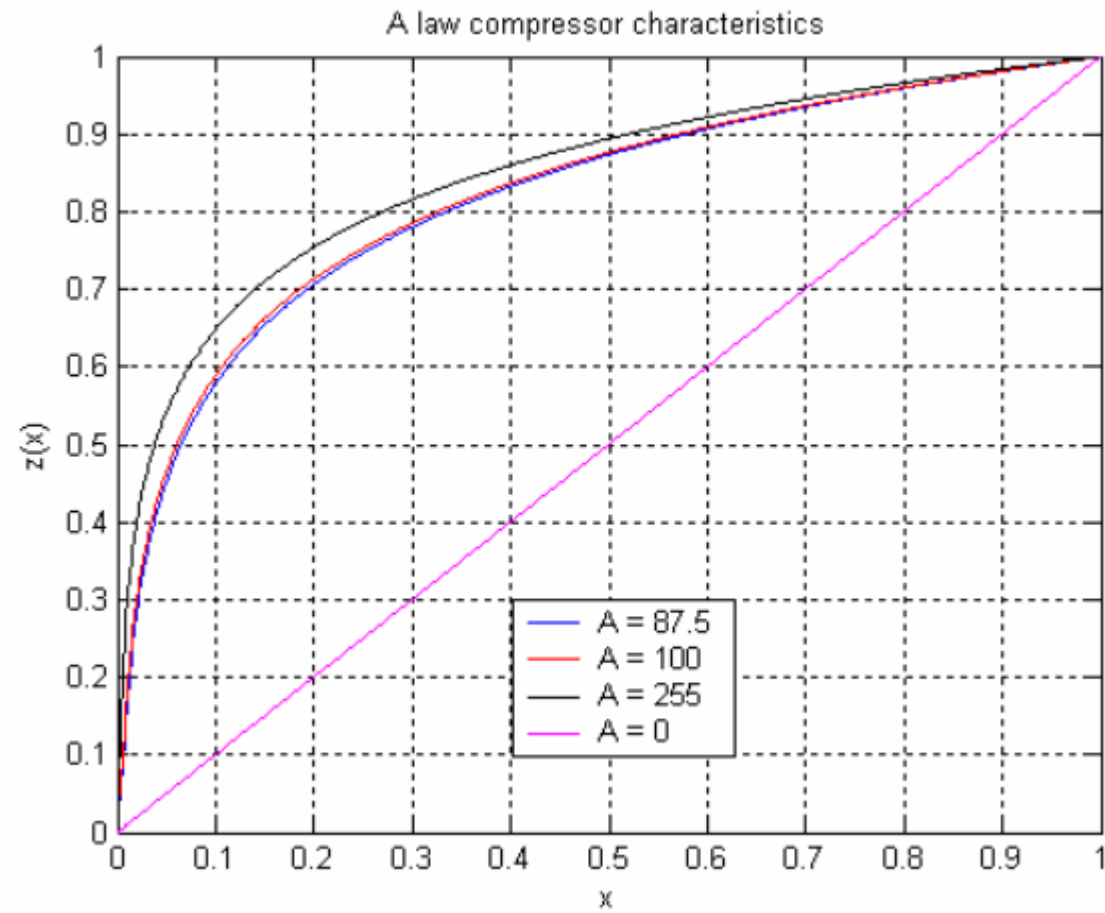


A - law companding

$$\frac{c(|x|)}{V} = \frac{A \frac{|x|}{V}}{1 + \ln A}, \quad 0 \leq \frac{|x|}{V} \leq \frac{1}{A}$$
$$= \frac{1 + \ln \left(A \frac{|x|}{V} \right)}{1 + \ln A}, \quad \frac{1}{A} \leq \frac{|x|}{V} \leq 1.0$$

Typical Value used in practice is $A=87.5$





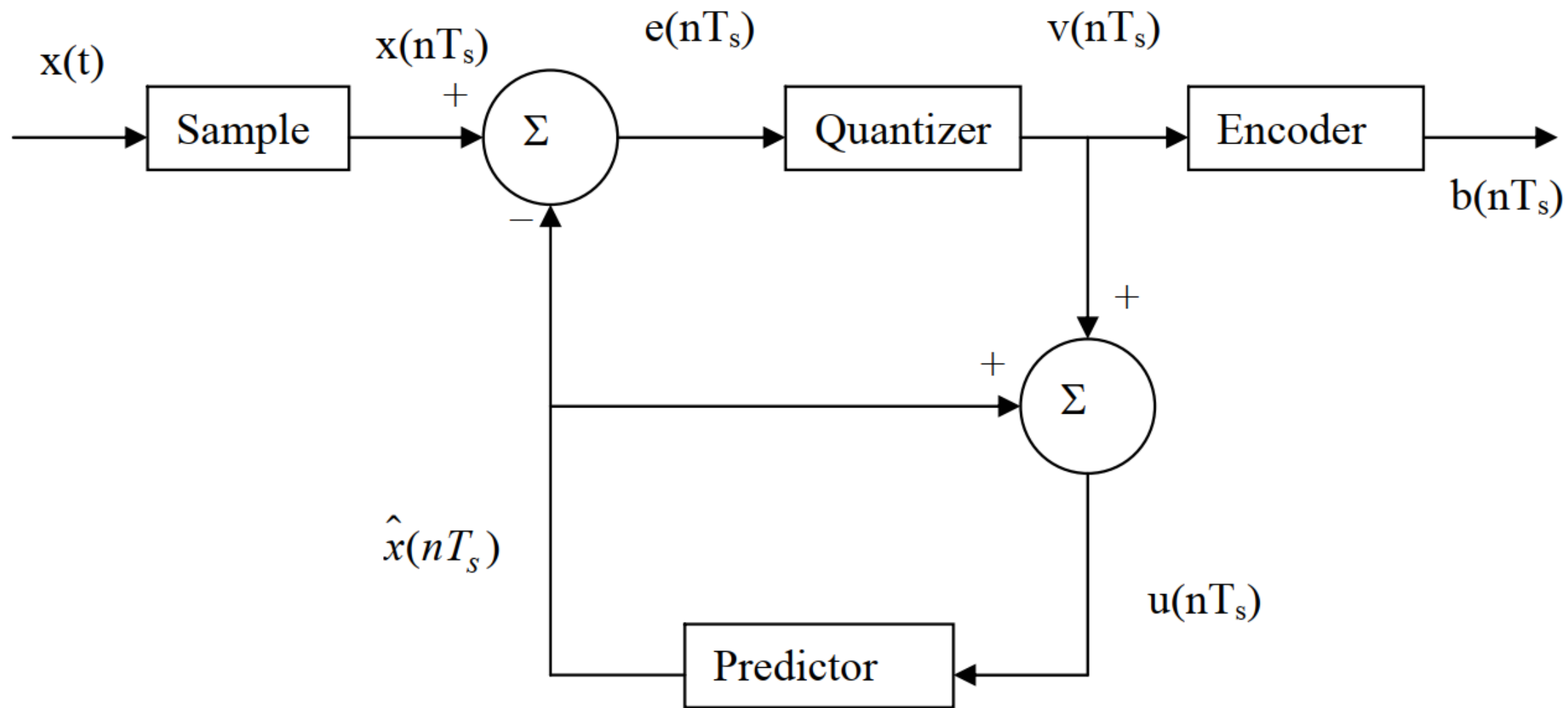


Fig. 3.13.2 *Schematic diagram of a DPCM modulator*



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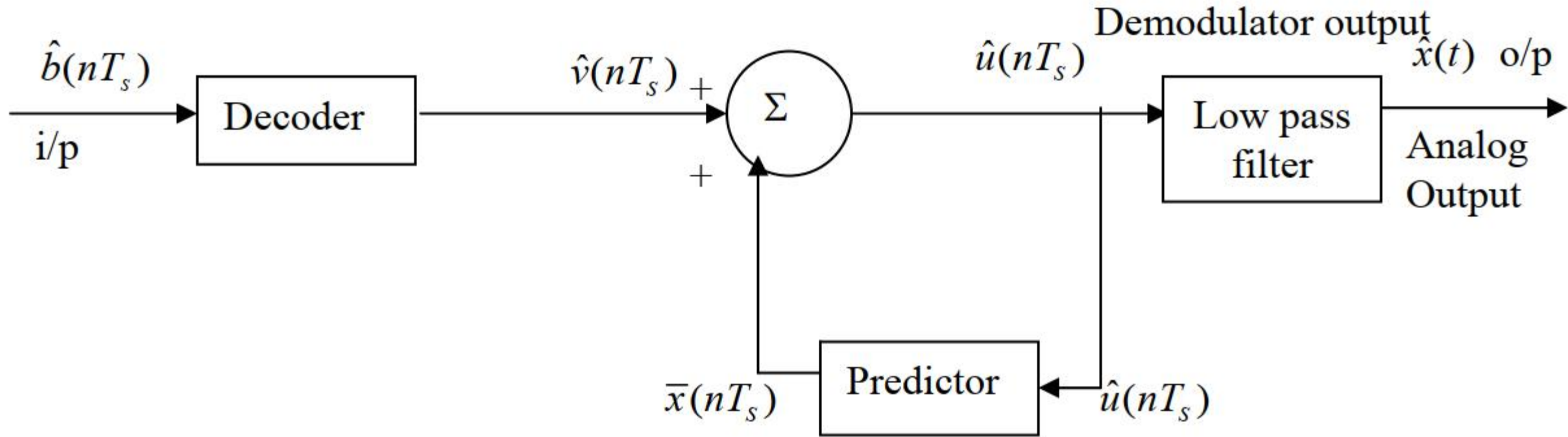


Fig. 3.13.3 Schematic diagram of a DPCM demodulator; note that the demodulator is very similar to a portion of the modulator



A message signal $m(t)$ is sampled and transmitted through a channel using 4-bit PCM. The Sampling rate is 50% higher than the Nyquist rate. Calculate the parameters of PCM.

$$m(t) = 10 \cos(4\pi 10^3 t)$$



A video signal is band limited to 4.5 MHz and transmitted through a channel. Determine the sampling rate if the signal is sampled at least 20% higher than the Nyquist rate. Determine the data rate of the PCM system if there are 1024 quantization levels.



The message signal is transmitted through a channel using PCM the sampling rate is twice the Nyquist rate, the minimum SQNR should be 22 dB. Determine the bit rate, Signal power, quantization power and SQNR.

$$m(t) = 5 \cos(2\pi 10^3 t)$$





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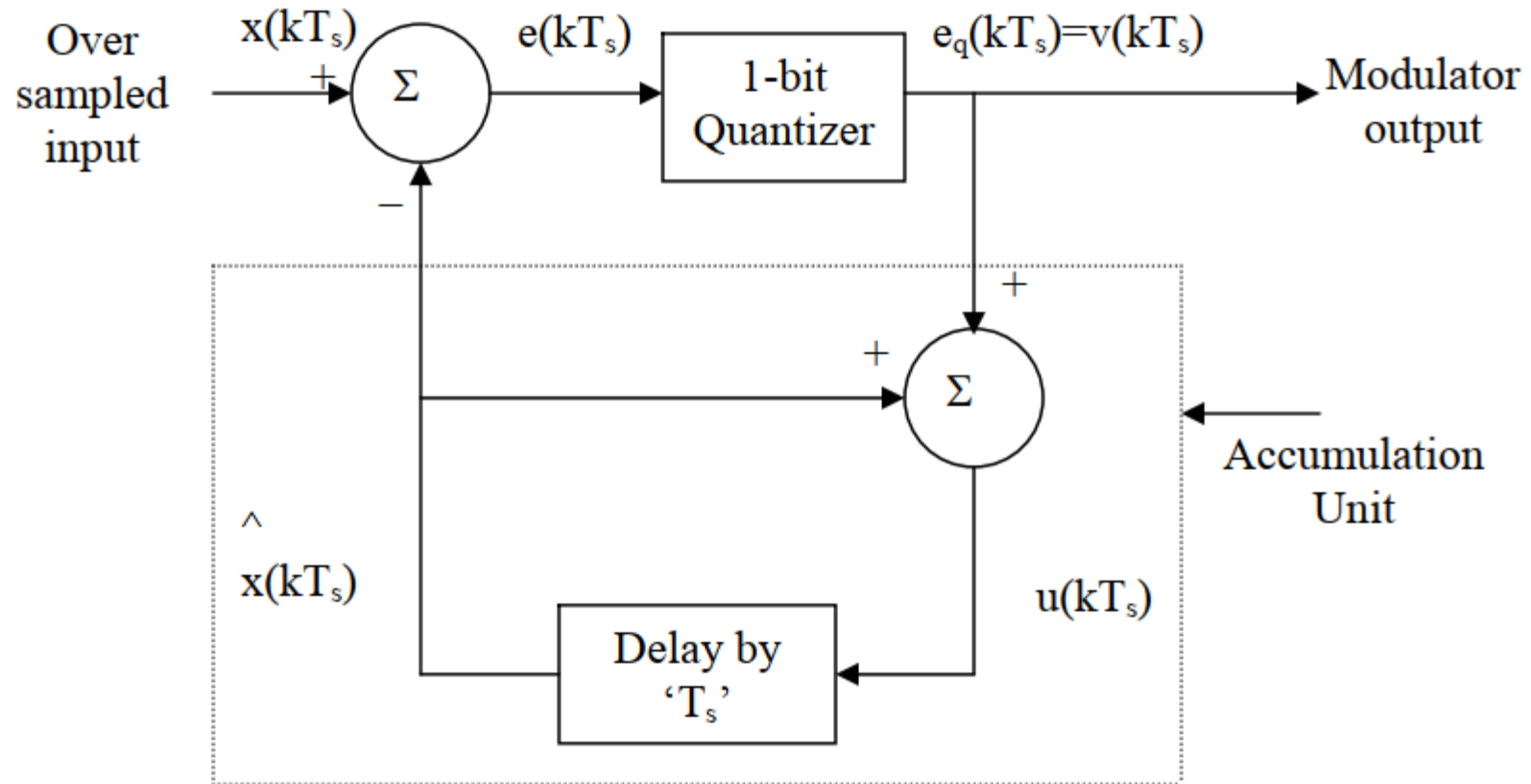


Fig. 3.14.2 Block diagram of a delta modulator



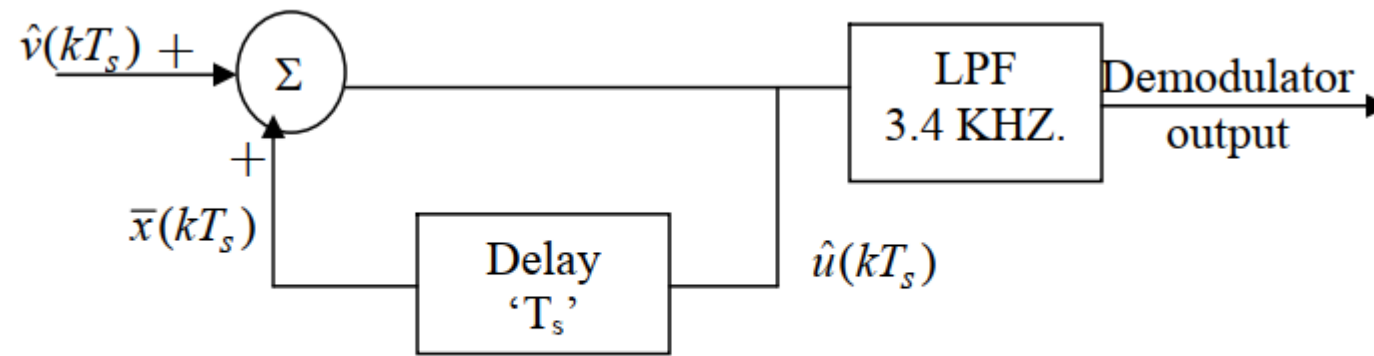
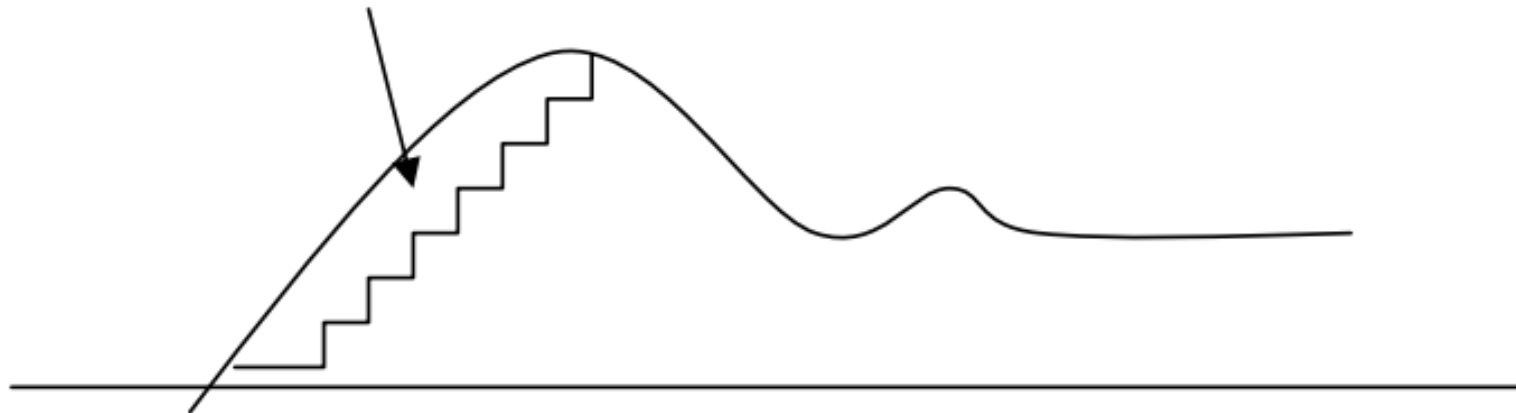


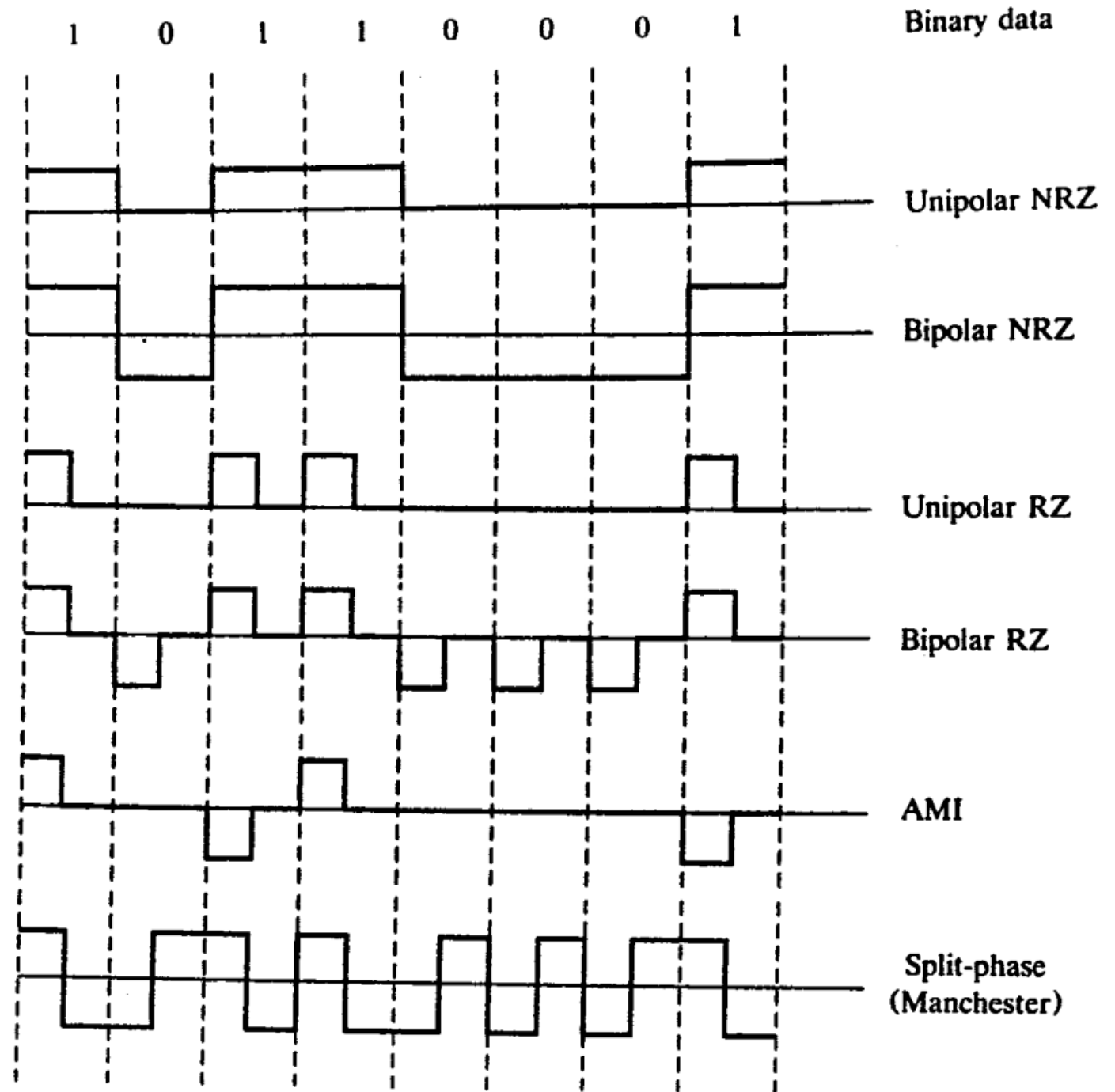
Fig. 3.14.4 *Demodulator structure for DM*



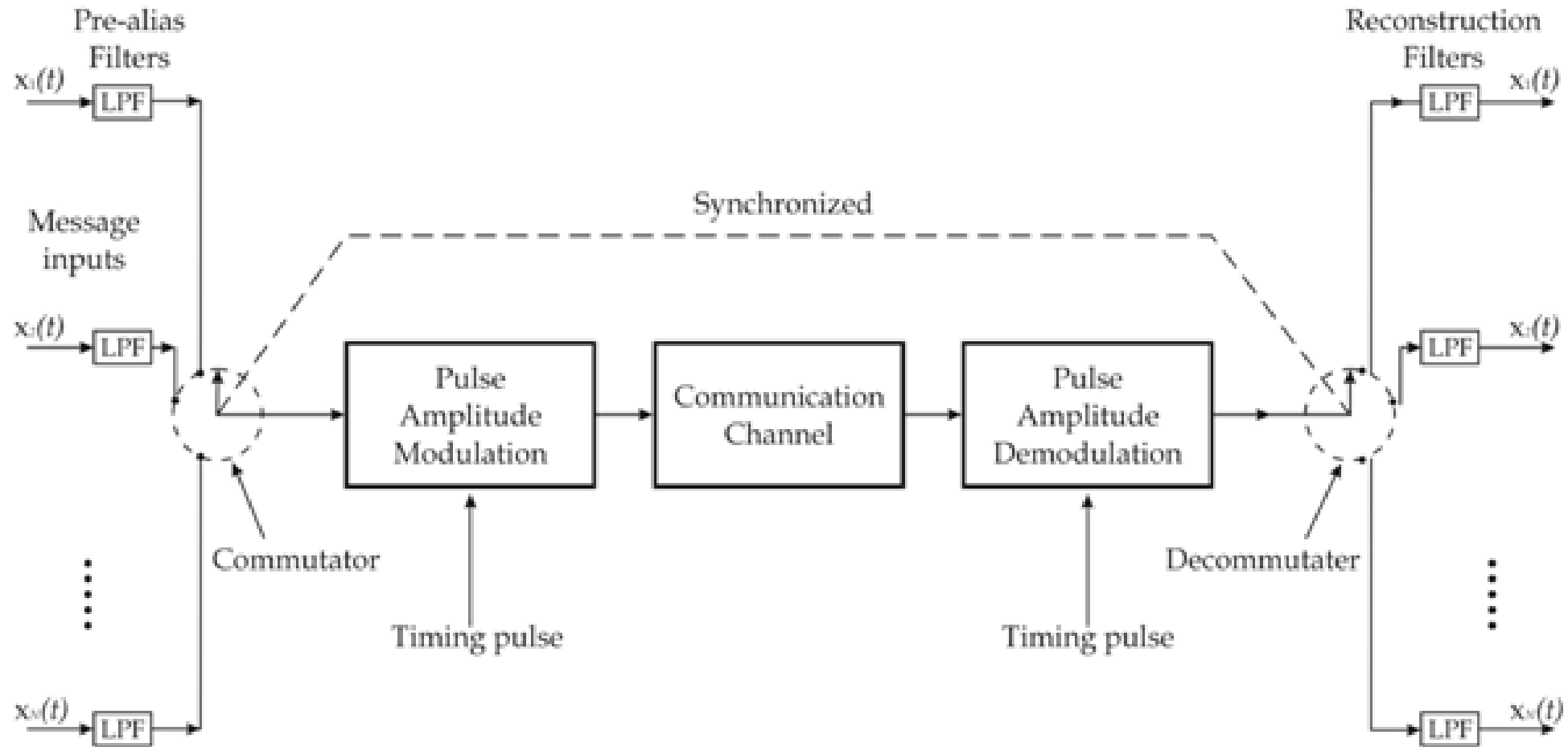
Slope over load
distortion



Signaling Formats



TIME DIVISION MULTIPLEXING (TDM)



TIME DIVISION MULTIPLEXING (TDM)

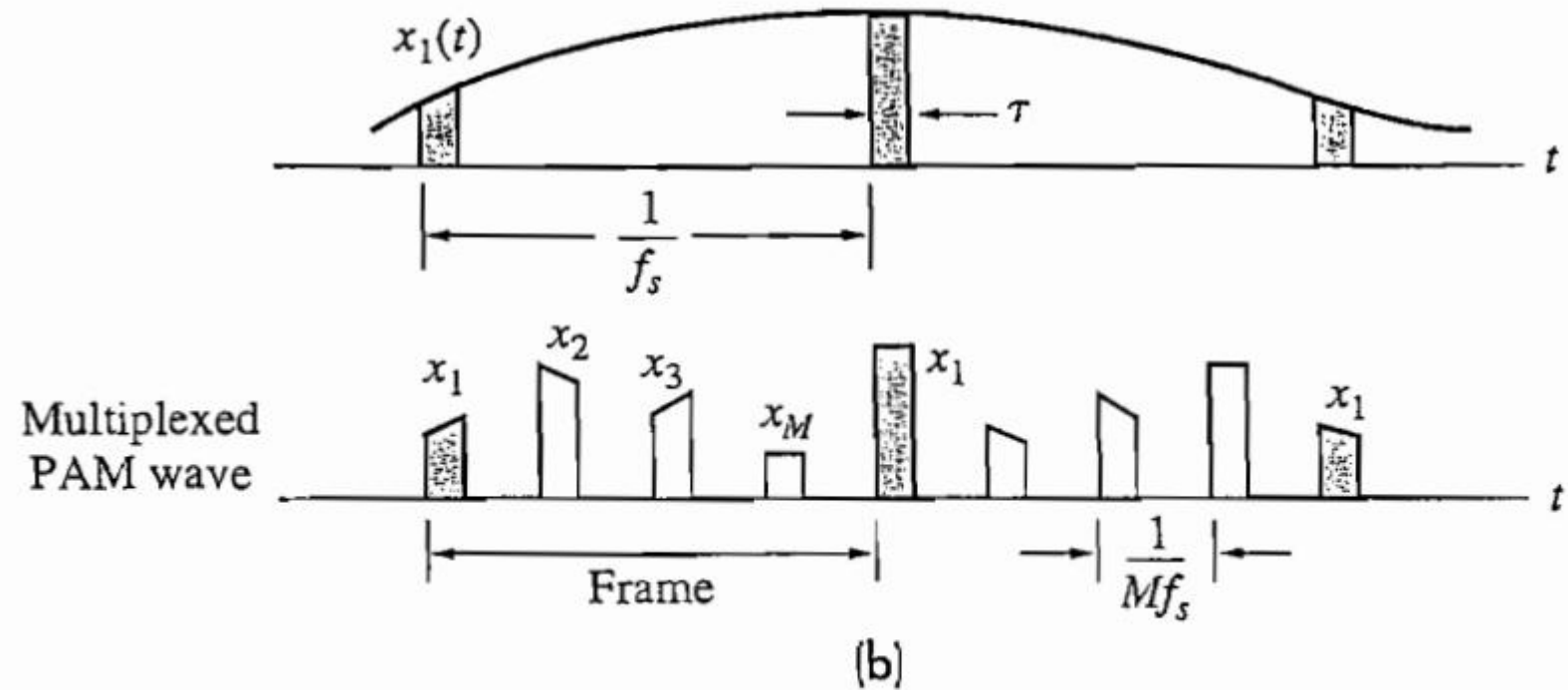
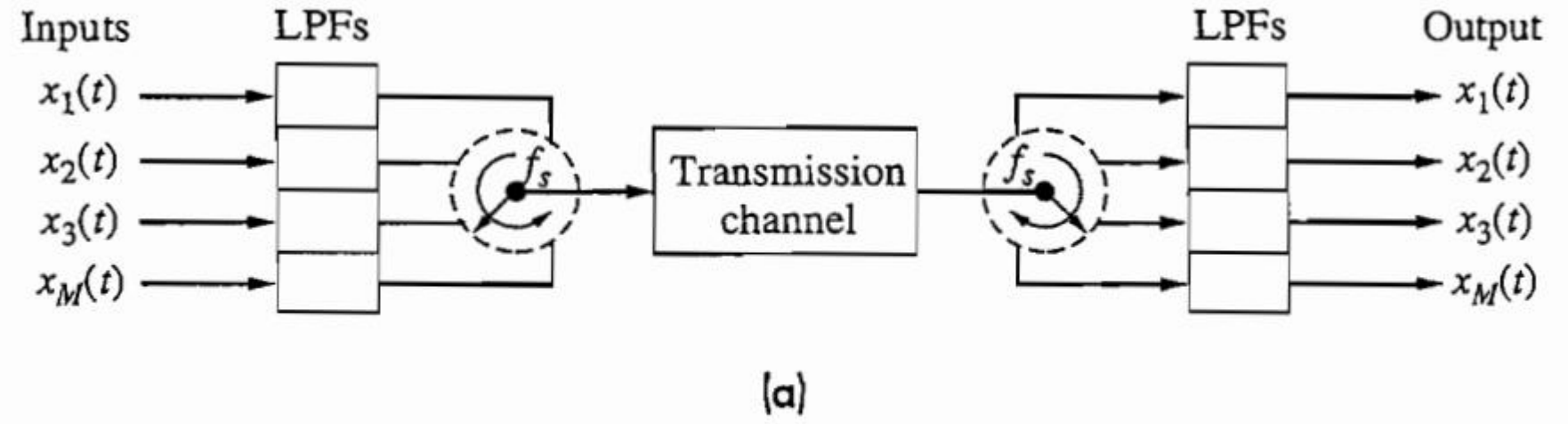


Figure 7.2-7 TDM system. (a) Block diagram; (b) waveforms.

- TDM used for multiplexing digital signals.
- Time taken by commutator to complete one rotation is called as frame time T_s .
- LPF is anti aliasing filter.
- The commutator is a rotating switch which rotates in the anti clock wise direction with uniform speed.
- Decommutator is also a rotating switch which rotates in clockwise direction and separates each of the message signal and produce at i/p of LPF.
- If N - no. of multiplexed signals
n - no. of bits / samples then

$$\text{Frame time} = T_s = N.n.T_b$$

$$\text{Bit duration} = T_b = \frac{T_s}{N.n}$$

$$\text{Bit Rate} = R_b = \frac{1}{T_b} = n.N.f_s$$



- To maintain synchronization between commutator and decommutator synchronization bits(a) are sent after each message signal.
- So, If synchronization bits present. then

$$(nN + a)T_b = T_s$$

$$T_b = \frac{T_s}{nN + a}$$

$$R_b = (nN + a)f$$



Differences between TDM and FDM

- In FDM carrier frequencies will be unique for message signals to be multiplexed but in TDM time slots are unique.
- In FDM sharing of channel band width will be done where as in TDM no sharing of band width.
- FDM is complex than TDM.
- The effect of Noise will be high in FDM as compared to TDM.



A message to a DM is $6\sin 2\pi 10^3 t$. Determine

- (a) The pulse rate that will prevent the error if the step size is $0.314V$
- (b) If the pulse rate for the above system is $56,000$ samples/sec. Find the value of the step size that will avoid distortion.



Consider the binary sequence 0100101. Draw the waveforms for the following signaling formats.

- (a) Unipolar NRZ signaling format
- (b) Bipolar RZ signaling format
- (c) AMI (alternate mark inversion) RZ signaling format



The input to a linear delta modulator having a step-size $\Delta = 0.628$ is a sine wave with frequency f_m and peak amplitude E_m . If the sampling frequency $f_s = 40$ kHz, the combination of the sine-wave frequency and the peak amplitude, where slope overload will take place is **(GATE-2003)**

E_m	f_m
(a) 0.3 V	8 kHz
(b) 1.5V	4kHz
(c) 1.5V	2kHz
(d) 3.0V	1kHz

A signal is sampled at 8 kHz and is quantized using 8-bit quantizer. Assuming SNR_q for a sinusoidal signal, the correct statement for PCM signal of bit rate of R is **(GATE-2003)**

- | | |
|--------------------------------------|--------------------------------------|
| (a) $R = 32$ kbps, $SNR_q = 25.8$ dB | (b) $R = 64$ kbps, $SNR_q = 49.8$ dB |
| (c) $R = 64$ kbps, $SNR_q = 55.8$ dB | (d) $R = 32$ kbps, $SNR_q = 49.8$ dB |

In a PCM system. if the code word length is increased from 6 to 8 bits, the signal to quantization noise ratio improves by the factor **(GATE-2004)**

- | | |
|---------|--------|
| (a) 8/6 | (b) 12 |
|---------|--------|

(c) 16

(d) 8



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Four Message signals each bandlimited to 5kHz are transmitted through a channel using TDM. The modulation used is 10-bit PCM. Determine the bit rate and bandwidth of the multiplexed signal.



Five voice signals each bandlimited to 4kHz are transmitted through a channel using TDM. The modulation used is 8-bit PCM. Time to complete one full revolution of commutator is 125 μ sec. Two extra bits are required for each frame to synchronize. Determine the bit rate and bandwidth of the multiplexed signal.





In a binary PCM system, the output signal-to-quantizing-noise ratio is to be held to a minimum of 40 dB. Determine the number of required levels, and find the corresponding output signal-to-quantizing-noise ratio.

The output *signal-to-quantizing-noise ratio* $(\text{SNR})_o$ in a PCM system is defined as the ratio of average signal power to average quantizing noise power. For a full-scale sinusoidal modulating signal with amplitude A , show that

$$(\text{SNR})_o = \left(\frac{S}{N_q} \right)_o = \frac{3}{2} L^2 \quad (5.46)$$

or

$$\left(\frac{S}{N_q} \right)_{0 \text{ dB}} = 10 \log \left(\frac{S}{N_q} \right)_o = 1.76 + 20 \log L \quad (5.47)$$

where L is the number of quantizing levels.

The Nyquist sampling frequency (in Hz) of a signal $6 \times 10^4 \text{sinc}^3(400t) \times 10^6 \text{sinc}^3(100t)$ is

(GATE-1999)

- a) 200 b) 300 c) 1500 d) 1000

The peak-to-peak input to an 8-bit PCM coder is 2 volts. The signal power-to-quantization noise power ratio (in dB) for an input of $0.5 \cos(\omega_m t)$ is

(GATE-1999)

- a) 47.8 b) 43.8 c) 95.6 d) 99.6

A video transmission system transmits 625 picture frames per second. Each frame consists of a 400X400 pixel grid with 64 intensity levels per pixel. The data rate of the system is

(GATE-2001)

- a) 16 Mbps b) 100Mbps c) 600Mbps d) 6.4 Gbps

Four independent messages have bandwidths of 100 Hz, 100 Hz, 200 Hz, and 400Hz respectively. Each is sampled at the Nyquist rate, and the samples are time division multiplexed and transmitted. The transmitted sample rate (in Hz) is

(GATE-1999)

- a) 1600 b) 800 c) 400 d) 200

